

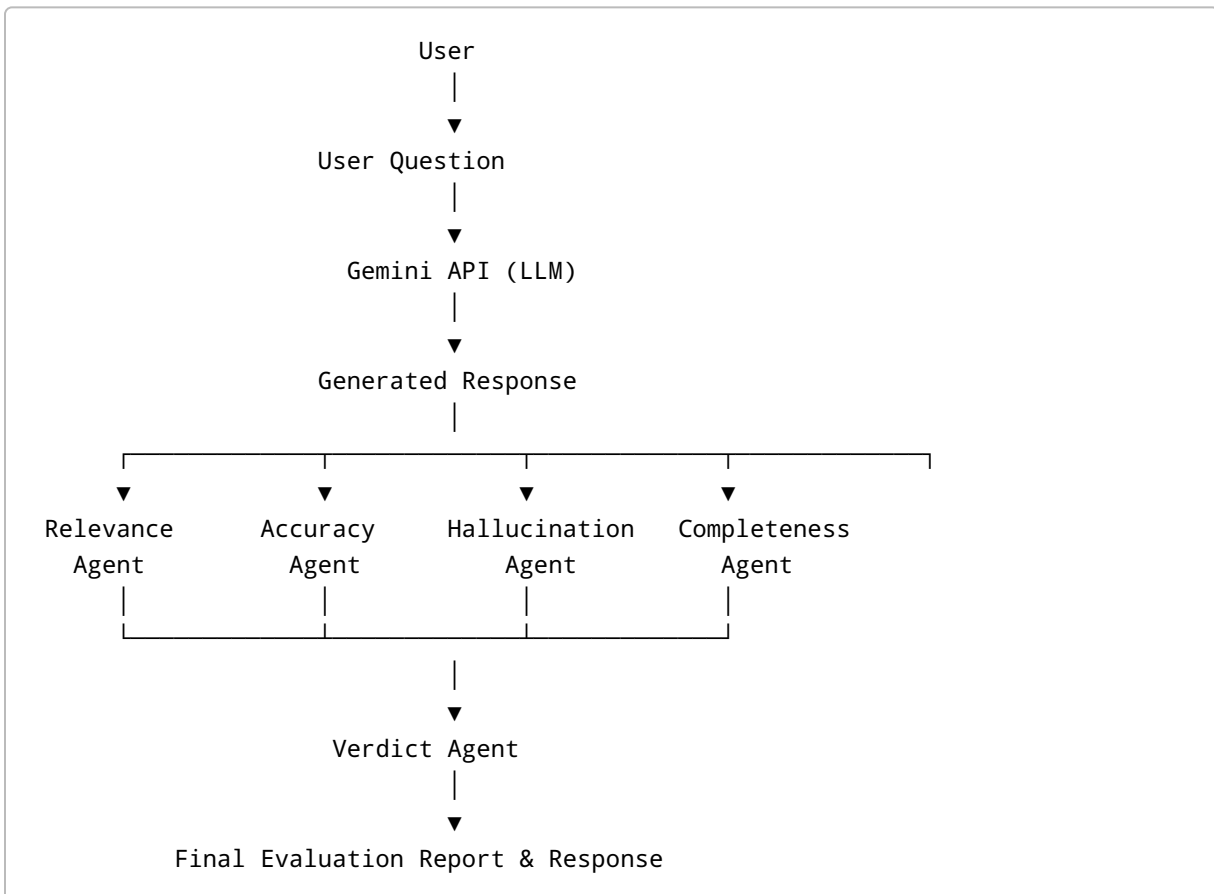
Agent Responsibilities

Introduction

The proposed AI evaluation system follows a **Multi-Agent Architecture** in which multiple specialized AI agents collaborate to evaluate the quality of responses generated by the Large Language Model (LLM). Instead of assigning all evaluation tasks to a single agent, the system distributes responsibilities among multiple independent agents. Each agent focuses on one quality aspect of the generated response, such as relevance, accuracy, hallucination detection, and completeness.

This modular architecture improves evaluation reliability, simplifies debugging, enables independent optimization of each evaluation criterion, and supports future scalability. After all agents complete their individual evaluations, a final **Verdict Agent** combines their outputs to generate the overall response quality score and recommendation.

Multi-Agent Evaluation Architecture



Overall Agent Workflow

1. The user submits a question through the application.
 2. The RAG pipeline retrieves relevant document chunks.
 3. Gemini generates a response using the retrieved context.
 4. The generated response is sent simultaneously to all evaluation agents.
 5. Each agent evaluates one quality dimension independently.
 6. The Verdict Agent collects all evaluation scores.
 7. The Verdict Agent calculates the final quality score and generates the final evaluation report.
 8. The verified response is displayed to the user.
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1. Relevance Agent

Purpose

The Relevance Agent verifies whether the generated response directly answers the user's question and whether the retrieved documents are relevant to the query.

Its objective is to ensure that the system does not provide unrelated or off-topic information.

Inputs

- User Question
 - Retrieved Context
 - Generated Response
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Outputs

- Relevance Score (0–1 or 0–100)
 - Relevant / Not Relevant
 - Feedback explaining the score
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Responsibilities

- Compare the question with the generated answer.
- Measure semantic similarity between the query and response.
- Verify that retrieved documents match the user's intent.
- Detect irrelevant or partially relevant responses.
- Generate a relevance confidence score.

Example

Question:

"What is Machine Learning?"

Generated Answer:

"Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data."

Result:

Relevance Score = 0.98

Status = Highly Relevant

2. Accuracy Agent

Purpose

The Accuracy Agent verifies whether the generated information is factually correct.

Instead of checking relevance, it compares the generated response with the retrieved reference documents to identify factual correctness.

Inputs

- User Question
 - Retrieved Documents
 - Generated Response
-

Outputs

- Accuracy Score
 - Correct / Incorrect
 - List of factual inconsistencies
-

Responsibilities

- Compare generated statements with retrieved documents.
 - Identify incorrect facts.
 - Detect outdated information.
 - Validate numerical values, names, dates, and definitions.
 - Measure factual correctness.
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Example

Retrieved Context:

"Python was created by Guido van Rossum."

Generated Answer:

"Python was created by Guido van Rossum."

Accuracy Score = 100%

Generated Answer:

"Python was developed by Microsoft."

Accuracy Score = Low

3. Hallucination Agent

Purpose

The Hallucination Agent identifies statements generated by the LLM that are unsupported by the retrieved documents or contain fabricated information.

This agent is responsible for improving AI trustworthiness.

Inputs

- Retrieved Context
- Generated Response

Outputs

- Hallucination Score
 - Hallucination Detected (Yes/No)
 - Unsupported Statements
-

Responsibilities

- Compare every generated statement with retrieved context.
 - Detect fabricated facts.
 - Detect fake citations.
 - Detect unsupported claims.
 - Measure response faithfulness.
 - Reduce misinformation.
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Example

Retrieved Document:

"Earth revolves around the Sun."

Generated Response:

"Earth revolves around Mars."

Result:

Hallucination Detected = Yes

Faithfulness = Low

4. Completeness Agent

Purpose

The Completeness Agent evaluates whether the generated response covers all important aspects of the user's question.

Even if an answer is correct, it may still be incomplete.

Inputs

- User Question
 - Retrieved Context
 - Generated Response
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Outputs

- Completeness Score
 - Missing Topics
 - Suggestions for improvement
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Responsibilities

- Identify important concepts expected in the answer.
 - Compare expected topics with generated content.
 - Detect missing explanations.
 - Measure answer coverage.
 - Improve response quality.
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Example

Question:

"Explain Machine Learning."

Expected Topics:

- Definition
- Types
- Applications
- Advantages

Generated Answer:

Only definition provided.

Completeness Score = 40%

Missing Topics:

- Types
 - Applications
 - Advantages
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5. Verdict Agent

Purpose

The Verdict Agent is the final decision-making component of the evaluation system.

It receives evaluation scores from all previous agents and generates the overall response quality assessment.

Inputs

- Relevance Score
 - Accuracy Score
 - Hallucination Score
 - Completeness Score
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Outputs

- Final Quality Score
 - Overall Grade
 - Final Recommendation
 - Detailed Evaluation Report
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Responsibilities

- Collect all evaluation results.
 - Calculate the overall quality score.
 - Assign response grade.
 - Generate improvement suggestions.
 - Produce the final evaluation report.
-

Example

Metric	Score
Relevance	95
Accuracy	92
Hallucination	98
Completeness	88

Final Quality Score = 93%

Overall Grade = Excellent

Recommendation:

"The generated response is factually accurate, relevant, and well grounded. Minor improvements can be made to increase completeness."

Agent Input and Output Summary

Agent	Inputs	Outputs
Relevance Agent	User Question, Retrieved Context, Response	Relevance Score, Feedback
Accuracy Agent	Retrieved Context, Response	Accuracy Score, Fact Verification
Hallucination Agent	Retrieved Context, Response	Hallucination Score, Unsupported Claims
Completeness Agent	User Question, Retrieved Context, Response	Completeness Score, Missing Topics
Verdict Agent	All Agent Scores	Final Score, Grade, Recommendation

Integration with RAGAS and TruLens

The proposed agents utilize the evaluation frameworks selected for the project.

RAGAS supports:

- Relevance Agent (Answer Relevancy)

- Hallucination Agent (Faithfulness)
- Completeness Agent (Context Recall)
- Relevance Agent (Context Precision)

TruLens supports:

- Response Logging
- Prompt Tracking
- Groundedness Evaluation
- Performance Monitoring
- Feedback Collection
- Dashboard Visualization

Together, RAGAS provides quality metrics, while TruLens records and monitors the complete evaluation process.

Advantages of the Multi-Agent Architecture

- Modular design with independent evaluation agents.
 - Easier debugging and maintenance.
 - Parallel execution of evaluation tasks.
 - Better hallucination detection.
 - Improved response quality.
 - Scalable architecture for future enhancements.
 - Easy integration with RAGAS and TruLens.
 - Supports enterprise-level AI evaluation systems.
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Conclusion

The proposed Multi-Agent Evaluation Architecture divides the responsibility of response evaluation among specialized agents, ensuring a comprehensive assessment of AI-generated content. The Relevance Agent verifies whether the response addresses the user's question, the Accuracy Agent checks factual correctness, the Hallucination Agent detects unsupported or fabricated information, and the Completeness Agent measures the coverage of the response. Finally, the Verdict Agent aggregates all evaluation scores to generate a final quality assessment and recommendation.

By integrating these agents with the RAG pipeline, RAGAS evaluation metrics, and TruLens monitoring capabilities, the proposed system provides a scalable, transparent, and reliable framework for evaluating Large Language Model responses while significantly reducing hallucinations and improving user trust.